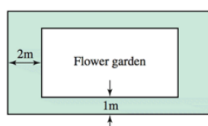


Problem 2

Problem

by a ξ
(see figure)
garden



- Label the picture with variables.
- What are you trying to maximize or minimize? Write an equation for it in terms of the variables from (a).
- What is your constraint? Write a constraint equation in terms of the variables from (a).
- Reduce your optimization equation to one variable using the constraint equation.
[2]
- What is the interval on which your variable makes sense? Is it open or closed? What does this mean for the method of finding the global max or min? [2]
- Use the appropriate method to find and justify your global extremum. [4]
- Be sure to answer the question asked in the original problem.